Customer Shopping Behavior Analysis

Project Overview

This project dives into customer shopping habits using a rich dataset of **10,000 purchases** spanning multiple product categories. By analyzing this data, our goal is to uncover key insights around:

- **Spending patterns** and how customers manage their budgets
- **Customer segmentation** to identify who your most valuable buyers are
- **Product preferences** to see which items drive the most interest
- **Subscription behavior** to understand what drives long-term loyalty

Ultimately, these insights are designed to help guide strategic business decisions and optimize future marketing and sales efforts.

Dataset Summary

Our analysis is built on a robust dataset spanning **10,000 purchase records** across **18 distinct columns**. To give you a clear picture of what we're working with, the data breaks down into three core categories:

- **Customer Demographics:** Tracks essential buyer details like age, gender, location, and whether they hold an active subscription status.
- **Purchase Details:** Captures specifics about what was bought, including the item name, product category, purchase amount, seasonal timing, size, and color.
- **Shopping Behavior:** Details the buyer's habits, such as whether a discount or promo code was applied, their past purchase history, how frequently they shop, their review ratings, and their preferred shipping method.

Data Cleanliness Note: Out of the entire dataset, we identified a minor gap with **95 missing values** located in the review rating column, which we handled during our data preparation phase.

Exploratory Data Analysis & Preparation using Python

We kicked off the project using Python to clean, prepare, and explore the raw dataset:

- **Data Loading:** Imported the transactional records seamlessly into a DataFrame using the **pandas** library.
- **Initial Exploration:** Inspected the core structure of the dataset using df.info() and generated a comprehensive statistical overview using .describe() to spot trends, check distributions, and identify data types.

Out[4]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Ship
count	10000.00000	10000.000000	10000	10000	10000	10000.000000	10000	10000	10000	10000	9905.000000	10000	10000	1
unique	NaN	NaN	2	25	4	NaN	50	4	20	4	NaN	2	6	
top	NaN	NaN	Male	T-shirt	Clothing	NaN	New Hampshire	M	Green	Summer	NaN	No	Credit Card	Ex
freq	NaN	NaN	6738	594	6070	NaN	223	4383	543	2555	NaN	7218	1709	
mean	5000.50000	43.859300	NaN	NaN	NaN	58.800700	NaN	NaN	NaN	NaN	3.719768	NaN	NaN	
std	2886.89568	13.941167	NaN	NaN	NaN	16.798059	NaN	NaN	NaN	NaN	0.558895	NaN	NaN	
min	1.00000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	
25%	2500.75000	34.000000	NaN	NaN	NaN	47.000000	NaN	NaN	NaN	NaN	3.300000	NaN	NaN	
50%	5000.50000	44.000000	NaN	NaN	NaN	59.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	
75%	7500.25000	54.000000	NaN	NaN	NaN	70.000000	NaN	NaN	NaN	NaN	4.100000	NaN	NaN	
max	10000.00000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	

```
<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             10000 non-null  int64
 1   Age                     10000 non-null  int64
 2   Gender                  10000 non-null  str
 3   Item Purchased          10000 non-null  str
 4   Category                10000 non-null  str
 5   Purchase Amount (USD)   10000 non-null  int64
 6   Location                10000 non-null  str
 7   Size                    10000 non-null  str
 8   Color                   10000 non-null  str
 9   Season                  10000 non-null  str
 10  Review Rating           9905 non-null   float64
 11  Subscription Status     10000 non-null  str
 12  Payment Method          10000 non-null  str
 13  Shipping Type           10000 non-null  str
 14  Discount Applied        10000 non-null  str
 15  Promo Code Used         10000 non-null  str
 16  Previous Purchases      10000 non-null  int64
 17  Frequency of Purchases  10000 non-null  str
dtypes: float64(1), int64(4), str(13)
memory usage: 1.4 MB
```

After our initial exploration, we moved into the core data preparation and structuring phase:

- **Missing Data Handling:** Checked for null values across the board and intelligently imputed missing records in the review_rating column using the median rating for each specific product category.
- **Column Standardization:** Cleaned and renamed all columns into consistent **snake_case** for improved readability and seamless documentation.
- **Feature Engineering:**
    - Created an age_group column by effectively binning customer ages.
    - Generated a purchase_frequency_days column derived from existing purchase data.
- **Data Consistency Check:** Evaluated whether discount_applied and promo_code_used were redundant, ultimately dropping promo_code_used to streamline the dataset.
- **Database Integration:** Connected our Python script to MySQL and loaded the fully cleaned DataFrame directly into the database to set up our upcoming SQL analysis.

Data Analysis using SQL (Business Transactions)

We carried out structured queries to dive deeper into customer behavior and uncover actionable commercial insights:

- **Revenue by Gender –** Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Female	192466
	Male	395541

- **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	5	80
	10	59
	12	67
	14	68
	19	63
	23	64
	24	76
	25	92
	28	81
	29	69
	30	88
	37	68
	44	80
	49	60
	50	82
	54	64
	56	64

- **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
▶	Shoes	3.93
	Scarf	3.92
	Pants	3.91
	Socks	3.9
	Handbag	3.89

- **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	rate
▶	Standard	58.40
	Express	60.77

- **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
▶	Yes	2782	58.92	163914
	No	7218	58.75	424093

- **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
▶	Shoes	46.86
	Hoodie	46.27
	Scarf	45.23
	Shorts	44.76
	Boots	44.72

- **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of customers
▶	Loyal	8000
	Returning	1807
	New	193

- **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Belt	382
	2	Accessories	Handbag	382
	3	Accessories	Backpack	375
	1	Clothing	T-shirt	594
	2	Clothing	Dress	582
	3	Clothing	Hoodie	577
	1	Footwear	Sneakers	219
	2	Footwear	Sandals	207
	3	Footwear	Shoes	207
	1	Outerwear	Jacket	123
	2	Outerwear	Coat	119

- **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

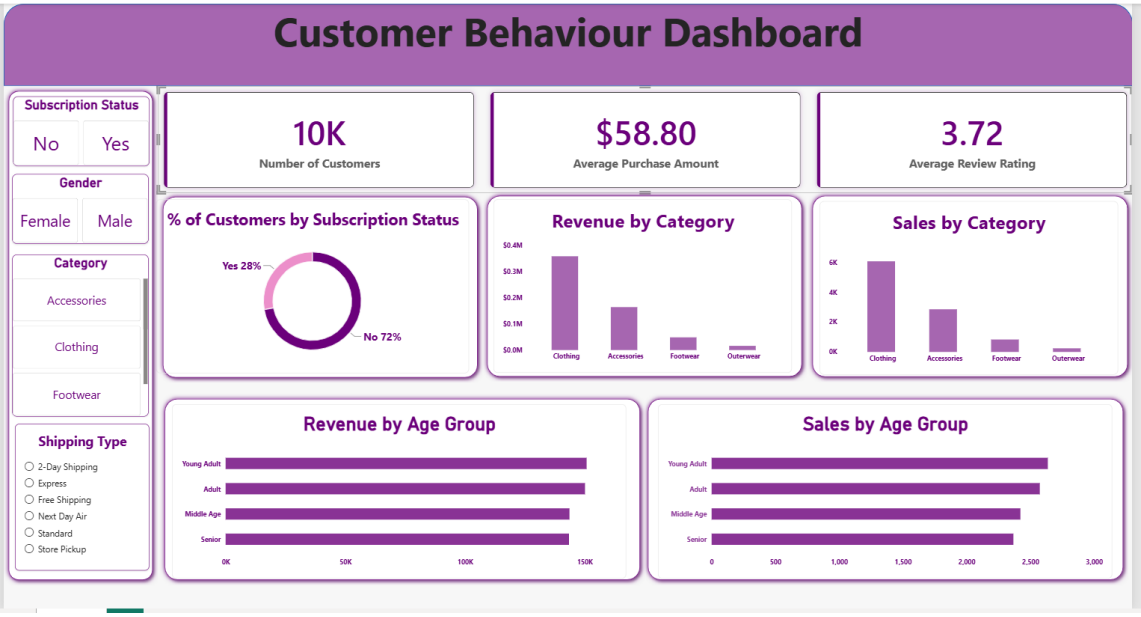
	subscription_status	repeat_buyers
▶	No	6411
	Yes	2572

- **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	Total revenue
▶	Young Adult	150796
	Adult	150125
	Middle Age	143647
	Senior	143439

Dashboard in Power BI

To bring all these findings to life, I built an interactive Power BI dashboard designed to present key metrics and customer insights visually at a glance.



Business Recommendation

1. Male customers represent a larger share of the customer base. The business should analyze purchasing behavior, spending patterns, and product preferences across both genders to identify potential growth opportunities and improve customer acquisition strategies.

2. The business should focus on increasing subscription adoption by highlighting exclusive benefits, offering limited-time subscription trials, and promoting subscription options during checkout. The goal is to increase the number of subscribers and overall revenue without reducing revenue per customer.
3. Develop targeted strategies for each customer segment. Encourage new customers to make a second purchase through personalized offers, motivate returning customers to purchase more frequently through loyalty rewards, and retain loyal customers by offering exclusive benefits and rewards.
4. Analyze the impact of discounts on sales volume, average order value, and profit margins. Adjust discount strategies to encourage more purchases while ensuring that increased sales translate into higher overall profitability.
5. Promote highly rated and best-selling products in marketing campaigns to attract customer attention, build trust, and potentially increase sales.
6. Focus marketing campaigns on age groups that generate the highest revenue and customers who prefer express shipping. Tailor promotions to their purchasing behavior to improve campaign effectiveness and increase revenue.